Hye-Young Jo

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Research Interests

I am a PhD student in Computer Science at the University of Colorado Boulder. My main research area is **Human-Computer Interaction**. My research goal is to **expand people's capacity to create and learn** by building interactive systems with generative AI and extended reality that adapt content to their actions, context, and intentions. My work spans AI-assisted video authoring, technology-enhanced learning, and embodied fitness training.

Keywords: human-AI interaction, creativity support tools, adaptive media, embodied interaction

Education

Summary: I trained in fine art, then industrial design, and now study computer science.

Aug 2024 - Present
University of Colorado Boulder
PhD student, Computer Science
Supervisor: Prof. Ryo Suzuki

Sep 2020 - Aug 2022
Korea Advanced Institute of Science and Technology (KAIST)
Master of Science, Industrial Design, *Best Master's Thesis Award*
Thesis: Exploring Different Augmented Reality Visualization Methods of Fitness Videos and Their Effects on the Exercise Experience
Supervisor: Prof. Andrea Bianchi

Mar 2011 - Feb 2016
Seoul National University
Bachelor of Fine Arts, Painting and Media Arts, *Cum Laude*
Painting graduation exhibition on fluid identity [A1]
Media Arts graduation exhibition on Korea's Sampo generation [A4]
Supervisor: Prof. Inhwan Oh, Prof. Cheol-Woong Sim

Research Experience

Summary: I have built interactive systems in five research groups across academia and industry, including research internships at Autodesk and Fujitsu.

May 2026 - Aug 2026
Autodesk, HCI and Visualization Team
Research Intern, Mentors: Dr. Frederik Brudy, Dr. David Ledo
Created a design space for film directing, grounded in a cinematic corpus, to establish a shared language between human creators and generative AI.

May 2025 - Oct 2025
Fujitsu Research of America, Converging Lab
Research Intern, Mentors: Dr. Mose Sakashita, Dr. Aditi Mishra, Dr. Aakar Gupta, Dr. Koichiro Niinuma
Built a system that grounds AI video generation in street-view imagery for spatially consistent shots, evaluated with 12 filmmakers [C7].

Aug 2024 - Present **University of Colorado Boulder, Programmable Reality Lab**
Graduate Research Assistant, Mentor: Prof. Ryo Suzuki
Study how people create and consume video by building interactive systems with generative AI, extended reality, and virtual agents for storytelling and education [C6, C7, R1].

Jan 2023 - Jul 2024 **Chung-Ang University, Human-AI Interaction Design & Fabrication Lab**
Research Assistant, Mentor: Prof. Yoonji Kim
Built a filmmaking previsualization tool using image segmentation and deepfake face swapping [C4], and haptic systems that recreate a personal trainer's touch through vibrotactile and electrical stimulation [E1, C5].

Sep 2020 - Sep 2022 **KAIST, Make Lab**
Graduate Research Assistant, Mentor: Prof. Andrea Bianchi
Designed and evaluated augmented reality (AR), virtual reality (VR), and haptic systems for home fitness, remote education, and immersive VR [C1-C3].

Work Experience

Summary: Before research, I spent four years at Dexter Studios, first as a film VFX compositor and later as a VR/AR generalist.

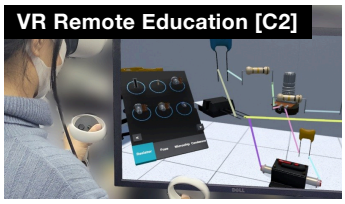
Aug 2018 - Apr 2020 **VR/AR Generalist at Dexter Studios**
New Media Division, Immersive Content and Virtual Production
Created 3D graphics for VR films, AR exhibitions and apps, and virtual production, covering character design, environments, visual scripting, lighting, and look development [X1-X2, X4].

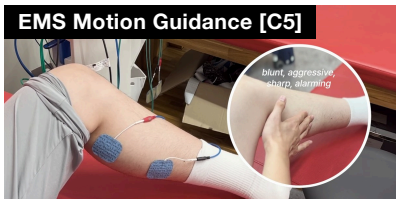
Sep 2016 - Jul 2018 **Film VFX Compositor at Dexter Studios**
Film Post-Production VFX Compositing Team
Composited live-action plates, 2D elements, and 3D CG renders into seamless final shots for feature films [F1-F6].

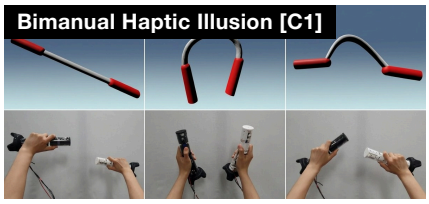
Research Overview

Summary: I have published 8 full papers in ACM CHI (4), ACM UIST (2), IEEE ISMAR (1, in IEEE TVCG), and HCI Korea (1), and have two more under review.









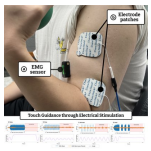
Peer-Reviewed Full Papers



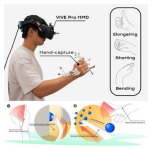
[C7] **Hye-Young Jo**, Mose Sakashita, Aditi Mishra, Ryo Suzuki, Koichiro Niinuma, Aakar Gupta. **Map2Video: Guiding Real-World-Grounded AI Video Generation**. In Proceedings of the Annual ACM Symposium on User Interface Software and Technology. ACM, 2026.
(**UIST '26**, acceptance rate: 20.1%) Also accepted to the UIST '26 demo track. (video) (PDF)



[C6] Keiichi Ihara, DaeHo Lee, Manato Abe, **Hye-Young Jo**, Ryo Suzuki. **CinemaWorld: Generative Augmented Reality with LLMs and 3D Scene Generation for Movie Augmentation**. In Proceedings of the Annual ACM Symposium on User Interface Software and Technology. ACM, 2026.
(**UIST '26**, acceptance rate: 20.1%) Also accepted to the UIST '26 demo track. (video) (PDF)



[C5] Dong-Uk Kim, **Hye-Young Jo**, Hankyung Kim, Ryo Suzuki, Seungwoo Je, Yoonji Kim. **TingleTouch: Touch Guidance through Electrical Stimulation in Resistance Training**. In Proceedings of the CHI Conference on Human Factors in Computing Systems. ACM, 2026.
(**CHI '26**, acceptance rate: 25.3%) (video) (PDF)



[J1] Yilong Lin, Han Shi, Weitao Jiang, Xuesong Zhang, **Hye-Young Jo**, Yoonji Kim, Seungwoo Je. **Designing Hand and Forearm Gestures to Control Virtual Forearm for User-Initiated Forearm Deformation**. IEEE Transactions on Visualization and Computer Graphics. IEEE, 2025.
(**ISMAR '25**, acceptance rate: 7.86%) (video) (PDF)



[C4] **Hye-Young Jo**, Ryo Suzuki, Yoonji Kim. **CollageVis: Rapid Previsualization Tool for Indie Filmmaking using Video Collages**. In Proceedings of the CHI Conference on Human Factors in Computing Systems. ACM, 2024.
(**CHI '24**, acceptance rate: 26.4%) (video) (PDF)



[C3] **Hye-Young Jo**, Laurenz Seidel, Michel Pahud, Mike Sinclair, Andrea Bianchi. **FlowAR: How Different Augmented Reality Visualizations of Online Fitness Videos Support Flow for At-Home Yoga Exercises**. In Proceedings of the CHI Conference on Human Factors in Computing Systems. ACM, 2023.
(**CHI '23**, acceptance rate: 27.6%) (video) (PDF)



[C2] **Hye-Young Jo**, Wooje Chang, Hoonjin Jung, Andrea Bianchi. **Design of Virtual Reality Application for Interaction Prototyping Remote Education**. In Proceedings of HCI Korea. The HCI Society of Korea, 2022.
(**HCIK '22**) **Best Paper Award** (video) (PDF)



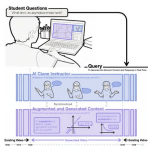
[C1] Neung Ryu, **Hye-Young Jo**, Michel Pahud, Mike Sinclair, Andrea Bianchi. **GamesBond: Bimanual Haptic Illusion of Physically Connected Objects for Immersive VR Using Grip Deformation**. In Proceedings of the CHI Conference on Human Factors in Computing Systems. ACM, 2021.
(**CHI '21**, acceptance rate: 26.3%) **Honorable Mention Award (top 5%)** (video) (PDF)

Posters



[E1] **Hye-Young Jo**, Chan Hu Wie, Yejin Jang, Dong-Uk Kim, Yurim Son, Yoonji Kim. **TrainerTap: Weightlifting Support System Prototype Simulating Personal Trainer's Tactile and Auditory Guidance**. In Adjunct Proceedings of the Annual ACM Symposium on User Interface Software and Technology. ACM, 2023.
(**UIST '23 Adjunct**, acceptance rate: 21.0%) (video) (PDF)

Preprints



[R1] **Hye-Young Jo**, Ada Zhao, Xiaoan Liu, Ryo Suzuki. **Generative Lecture: Making Lecture Videos Interactive with LLMs and AI Clone Instructors**. arXiv preprint. 2025.
(video) (PDF)

Fellowship and Awards

Summary: I have received $176,000 in fellowships, awards, and travel grants.

Fall 2025 - Spring 2027 **Google PhD Fellowship**
Awarded in the Human-Computer Interaction category; full tuition and stipend ($85,000/year).

2025 **Ralph J. Slutz Student Excellence Award**
Awarded by the Department of Computer Science, University of Colorado Boulder ($1,000).

2024 **Gary Marsden Travel Award**
Awarded by ACM SIGCHI to attend CHI 2024 ($3,000).

2024 - 2026 **Travel Grants, University of Colorado Boulder**
Conference Support Fund, Department of Computer Science ($1,500); Conference Travel Grant, Graduate and Professional Student Government ($500).

2023 **Best Master's Thesis Award**
Awarded to one of two master's theses in the Department of Industrial Design, KAIST.

2022 **Best Paper Award**
Awarded at HCI Korea 2022 [C2].

2021 **iF Design Award (Winner, UI/UX) and IDEA (Bronze, Digital Interaction)**
Contributed the information architecture to the control UI/UX of the Mobile Clinic Module, a negative-pressure isolation ward (team award, a multi-lab project of the Department of Industrial Design, KAIST).

2021 **Honorable Mention Award**
Awarded to the top 5% of submissions at CHI 2021 [C1].

Invited Talks and Lectures

Summary: I gave six talks and guest lectures on AI-assisted video creation, prototyping, and my transition from art and film VFX to immersive media at companies and universities in the U.S. and Korea.

Aug 2026 **Understanding Film Directing for AI-Assisted Video Creation, Autodesk Research**
Presented at the Summer Internship Showcase

Oct 2025 **Grounding Generative Video in Real-World Geographies for Spatial Consistency, Fujitsu Research of America**
Presented at a lunch seminar

May 2025 **Reimagining How We Create and Consume Content, KAIST and Seoul National University**
Presented at the Make Lab Seminar and the Human-Centered Computing Systems Lab Seminar

Feb 2025 **Utilizing AI Services and APIs for Coding and Virtual Agent Creation, University of Colorado Boulder**
Gave a guest lecture in Prof. Ryo Suzuki's How to Hack Almost Anything class (graduate course)

Oct 2022 **How to Make an Arduino Case with Laser Cutting (Fusion 360, Blender, Adobe Illustrator), KAIST**
Gave a guest lecture in Prof. Andrea Bianchi's Interaction Prototyping class (undergraduate course)

Nov 2019 **Transitioning from Art and Film VFX to VR Content Creation, Kookmin University**
Presented at the Department of Entertainment Design Career Seminar

Teaching Experience

Summary: I have taught introductory programming and physical prototyping as a teaching assistant.

Fall 2025 **TA for Introductory C++ Programming Class**
Worked as a teaching assistant in Prof. Rhonda Hoenigman's Starting Computing class, teaching introductory C++ programming at the University of Colorado Boulder

Fall 2022 **TA for Physical Computing Class**
Worked as a teaching assistant in Prof. Andrea Bianchi's Interaction Prototyping class, teaching physical computing with Arduino at KAIST

Service

2025 - Present **External Reviewer**
CHI 2025-2027, DIS 2026

Mentor

Spring 2024 Undergraduate research mentoring for Chan Hu Wie, Yejin Jang, Dong-Uk Kim, and Yurim Son (Chung-Ang University), guiding them through designing, building, and evaluating wearable haptic systems that recreate a personal trainer's touch, leading to a UIST poster [E1] and a CHI paper [C5]

2025 University of Colorado Boulder Prospect Affiliate Match Program, helping prospective students prepare their PhD applications

2012 Seoul National University Mentoring Program, helping international students adjust to campus life

Media Coverage

Oct 2025 **University of Colorado Boulder, Graduate School News**
"CU Boulder graduate student named a Google PhD fellow"

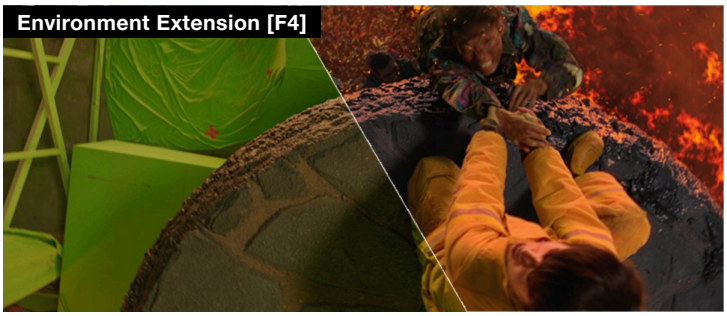
May 2021 **Microsoft Research Blog**
"Microsoft Research collaborates with KAIST in Korea to explore bimanual interactions with haptic feedback in virtual reality"

Film VFX (showreel)

Summary: I worked as a visual effects (VFX) compositor on six feature films, three of which each drew over five million viewers in South Korea.







2018 [F6] <Along with the Gods: The Last 49 Days> (VFX showreel), Director: Yong-Hwa Kim. *Best Visual Effects, 2018 Grand Bell Awards. Over 12 million viewers in South Korea as of 2024.*

2017 [F5] <1987: When the Day Comes> (VFX showreel), Director: Jun-Hwan Jang. *Best Film, 2018 Blue Dragon Film Awards. Over 5 million viewers in South Korea as of 2024.*

2017 [F4] <Along with the Gods: The Two Worlds> (VFX showreel), Director: Yong-Hwa Kim. *Four awards including Best Technical Achievement, 2018 Blue Dragon Film Awards. Over 11 million viewers in South Korea as of 2024.*

2017 [F3] <Kung Fu Yoga> (VFX showreel), Director: Stanley Tong

2017 [F2] <Real> (trailer), Director: Sa-Rang Lee

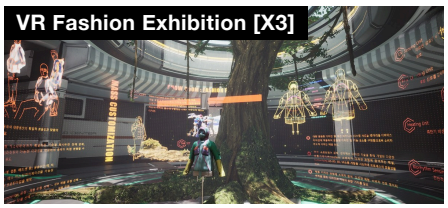
2017 [F1] <Fabricated City> (trailer), Director: Kwang-Hyun Park

Immersive Content (showreel)

Summary: I created immersive experiences across VR and AR, including a VR horror film and AR exhibition, an AR app featuring miniature K-pop idols, a commissioned VR showroom for a tech-fashion exhibition, and a VR boxing game. I worked across concept design, 3D modeling, lighting, look development, coding, and volumetric multi-camera capture.







2021 [X5] VR Boxing Game <Meta-Boxing> (video), Supervisor: Prof. Woontack Woo

2019 [X4] AR Mobile App <LGU+ 5G AR> (video, article), Supervisor: Sun-Gu Kim

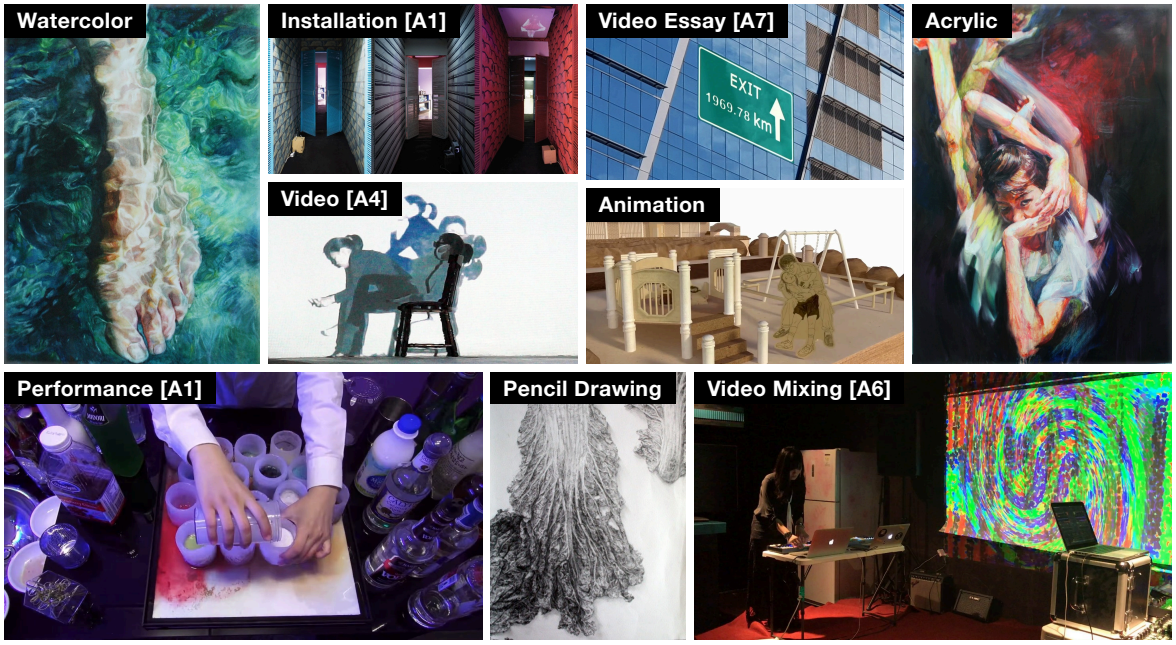
2019 [X3] VR Exhibition <Fashion For Help> (video), Supervisor: Young-Mo Son

2019 [X2] AR Exhibition <The Tide>, Supervisor: Sang-Hyoun Lee

2019 [X1] VR Toon Film <The Tide> (Steam), Director: Tae-Kyung Yoo. *Official selection, New Frontier, 2019 Sundance Film Festival.*

Art Exhibitions and Live Performances (showreel)

Summary: My art explores the human body, emotion, and relationships between people through identity and belonging, the precarity of young adults in Korea, and fear and isolation. Across painting, video, installation, and performance, I am drawn to real-time and participatory work that audiences help complete.



2018 [A7] Group Exhibition <Lapses> (video: Measurer), Platform-L Contemporary Art Center, Curator: Eobchae (funded by Hyundai's ZER01NE project)

2016 [A6] Live Video Jockey Performance <We Play>, Sangsangmadang, Supervisor: Hoon-Gyu Park (Parkpunk)

2016 [A5] Invited Exhibition <Don't fake it, believe it>, Mythtake Museum

2015 [A4] Group Exhibition <Dirt Luv for Graduation> (videos: Package for Me, Human-furniture, Walking on the Spot), Seoul National University

2015 [A3] Group Exhibition <8-bit>, Seoul National University

2015 [A2] Group Exhibition <Crawling>, Seoul National University

2014 [A1] Group Exhibition <The Great Exodus> (videos: Closet Inside the Closet, 1:1 Bar), Seoul National University. *One of three finalists for the Brighton Prize.*